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Annual Review of Virology

Heterogeneity and Collective Interactions Within Viral Populations

Mireille Farjo[1] and Christopher B. Brooke[1,2]

[1]Department of Microbiology, University of Illinois Urbana-Champaign, Urbana, Illinois, USA; email: cbrooke@illinois.edu

[2]Carl R. Woese Institute for Genomic Biology, University of Illinois Urbana-Champaign, Urbana, Illinois, USA

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Annu. Rev. Virol. 2026. 13:47–65

First published as a Review in Advance on May 26, 2026

The *Annual Review of Virology* is online at virology.annualreviews.org

https://doi.org/10.1146/annurev-virology-092623-093749

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Keywords

virus, heterogeneity, virus-virus interactions

Abstract

The existence of viruses as obligate intracellular parasites with high mutation rates sets them apart from other forms of life in fundamental ways. Frequent errors in replication and assembly processes give rise to substantial genetic and genomic diversity within viral populations. The diverse constituents of viral populations often interact within and across infected cells, giving rise to complex emergent phenotypes and genetic interactions. This review examines the causes and consequences of viral population diversity and explores how the complex collective interactions among viral genomes in cells and tissues blur the definition of individuality and influence patterns of viral replication and evolution across multiple biological scales.

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INTRODUCTION

Viral populations are enormously diverse due to high mutation rates and imperfections in the biochemical processes involved in genome replication, packaging, and assembly. While our technical ability to measure and account for this diversity has dramatically improved in recent years, the mechanistic origins and evolutionary consequences of different forms of population heterogeneity remain mysterious and controversial. Viral genomes and their products often interact in complex ways within infected cells and tissues. The prevalence of virus-virus interactions within infected cells, tissues, and organisms means that viral populations function more as collectives than as groups of independent individuals. Thus, viral population dynamics are governed by (*a*) the genetic and phenotypic heterogeneity present within the population and (*b*) how the diverse members of the population interact with each other across different levels of spatial and biological organization. Understanding these interactions and their consequences is critical for understanding how the replication and evolution of viral populations are influenced by different forms of viral diversity.

In this review, we describe the different forms of diversity and heterogeneity within viral populations and discuss their mechanistic origins and functional consequences. We then examine how the unique biology of viruses, as obligate intracellular parasites, gives rise to complex interactions among the diverse constituents of a population within infected cells that drive emergent behaviors and blur the definition of the individual. Altogether, we provide numerous examples from diverse viral systems that demonstrate how population heterogeneity and collective virus-virus interactions are unique features of viral biology central to understanding their replication and evolution.

DIVERSITY AND HETEROGENEITY WITHIN VIRAL POPULATIONS

Within a viral population, no two virions are the same. Individual virions typically exhibit significant diversity in genome sequence, genome structure, and morphology [reviewed elsewhere (1)]. This diversity both influences the population's collective behavior and serves as the raw material for evolution. Here, we aim to describe the different types of genetic and genomic heterogeneity commonly observed in viral populations (**Figure 1**) and to review what is known about the mechanistic underpinnings and functional consequences of this heterogeneity.

Genetic Diversity

Viral populations typically contain an enormous amount of standing genetic diversity (i.e., nucleotide substitutions) and are best understood as networks of related sequence variants (2). This diversity is continuously replenished through mutation and recombination and shaped by selection and genetic drift, as described below.

Mutation frequencies represent the observed frequencies of mutants within a population and are the product of the intrinsic mutation rate of the viral replicase [which is challenging to measure directly (3)] and the evolutionary forces that influence mutant accumulation/persistence. While mutation frequencies have long been known to vary between virus families and even between individual viral variants (4–6), only recent advances in sequencing approaches (7, 8) have made it possible to directly quantify mutation frequencies within viral populations. In general, mutation frequencies tend to be higher for RNA viruses compared with DNA viruses, but there may be some exceptions [e.g., some single-stranded DNA viruses exhibit high evolutionary rates suggestive of high mutation rates (9–12)]. The specific patterns likely differ between virus families due to differences in (*a*) intrinsic polymerase fidelity/mutation rates, (*b*) recombination rates (13), (*c*) the presence or absence of error correction (14), (*d*) effects of the host environment (15, 16),

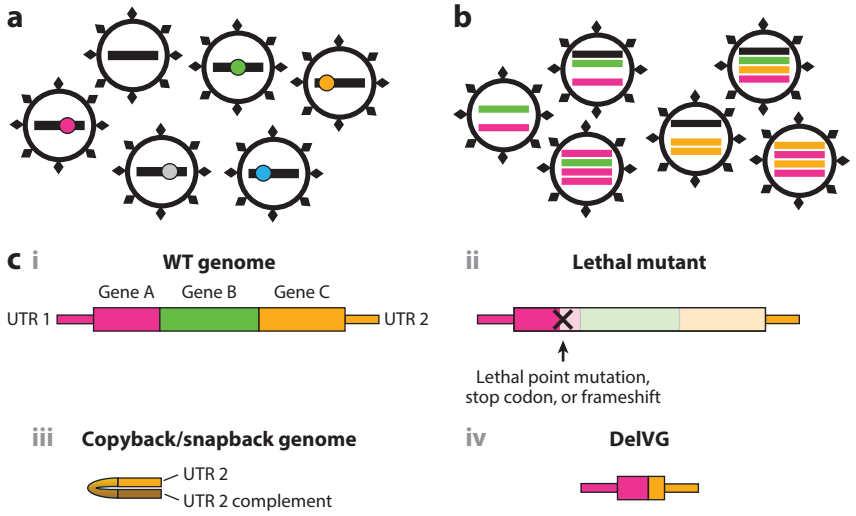


Figure 1

Forms of genetic and genomic heterogeneity. (*a*) Random point mutations arising from polymerase errors or cell-mediated mutagenesis. (*b*) Semi-infectious particles and other forms of gene packaging variability. (*c*) Different forms of NsVGs derived from a hypothetical WT genome: (*c*, *i*) WT virus genome. Colored boxes indicate individual open reading frames and UTR sequences as indicated. (*c*, *ii*) Viral genome carrying one or more lethal point mutations or frameshift mutations. An example of this is the DENV1 lineage encoding a truncated E protein described in References 63 and 64. (*c*, *iii*) Copyback and snapback genomes. (*c*, *iv*) DelVG containing a large, internal deletion. Abbreviations: DelVG, deletion-containing viral genome; DENV1, dengue virus type 1; NsVG, nonstandard viral genome; UTR, untranslated region; WT, wild type.

and (*e*) the degree of mutational robustness (17–21). Mutation frequencies also may vary across a given viral genome due to *cis*- and *trans*-acting factors that govern replicase function (22–25).

Interestingly, mutation rates may be partly controlled by pleiotropic alleles that also affect replicase speed, as observed with poliovirus, vesicular stomatitis virus (VSV), and human immunodeficiency virus type 1 (26–28). This suggests that, at least for some viruses, mutation-rate phenotypes may be linked to other viral traits through trade-offs, in which selection on other traits may constrain or influence polymerase-fidelity phenotypes.

The observed frequencies of variants within a viral population result not only from the rate at which they are introduced into the population through biochemical processes but also from how selection and genetic drift act on those mutations. Purifying selection can minimize the accumulation of deleterious mutations (29, 30). In contrast, genetic drift, which can dominate selection when within- or between-host transmission bottlenecks are narrow (31, 32) or when viral populations are highly spatially structured within a host (33–36), can result in the emergence or fixation of low-fitness variants. Thus, the relative balance of drift and selection acting on a population can be a major determinant of the amount of genetic diversity observed.

While mutation frequencies can clearly differ widely between viruses, the consequences of this variation are not always easy to predict and likely vary depending on the specific biology and life history of the virus in question. Certainly, higher levels of genetic diversity can facilitate broader exploration of sequence networks and increase the probability of accessing rare genotypes that can enhance fitness in the current environment or facilitate survival following environmental change. Obvious examples include immune escape variants and host-range variants. On the other hand, most random mutations are deleterious (22, 37–40); as a result, higher mutation rates should

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reduce the mean fitness of the population. Additionally, high deleterious mutation loads can reduce the virus's ability to adapt to new environments by limiting the success of rare beneficial alleles that would otherwise increase in frequency if they were not linked to deleterious mutations (41).

Genomic Diversity

Genomic diversity differs from genetic diversity in that it captures structural variation across packaged genomes. As detailed below, forms of genomic diversity include the products of recombination, insertions, deletions, and genome packaging variability. The specific modes and relative extent of genomic diversity remain poorly described for many viruses, but they appear to differ among virus families based on genome type, genome structure, replication strategy, and packaging mechanism. A common theme that unites different forms of genomic diversity is the potential to alter viral gene expression patterns through copy-number variation or the creation of new gene products.

Nonstandard viral genomes, deletion-containing viral genomes, and defective viral genomes. Even in the earliest days of experimental virology, it was observed that different viruses tended to exhibit dramatic increases in their particle-to-infectivity ratio when propagated under high multiplicity of infection (MOI) conditions (42–47). Researchers subsequently observed that a wide array of viruses commonly produce virions carrying genomes with major structural alterations, such as large deletions [deletion-containing viral genomes (DelVGs)] or copyback formations, that prevent the production of one or more essential viral genes (48–52). They were historically referred to as defective viral genomes (DVGs) because they can interfere with the replication of wild-type (WT) genomes under some conditions. However, the diverse array of structural variants commonly produced and packaged into virions is best captured by the umbrella term nonstandard viral genomes (NsVGs) (53). This term clearly distinguishes them from WT genomes without implying any further conclusions about their functionality, which remains poorly understood in most cases.

While the specific molecular mechanisms that give rise to NsVGs across different viruses remain murky (54, 55), it is clear that their production can significantly affect how viral populations replicate and interact with the host. This is because NsVGs, although often not replication competent on their own, remain biologically active. Depending on the nature of the deletion or structural rearrangement, NsVGs can encode new protein products distinct from those encoded by WT genomes. For instance, influenza virus DelVGs encode polypeptides that can act as dominant negative inhibitors of the viral polymerase, suggesting a possible role for DelVGs in modulating viral replicase function (56). NsVGs also can mediate unique effects on the host cell, including promoting cell survival and viral persistence, modulating viral assembly and translation, and enhancing activation of innate immune defenses (57–62). NsVGs can be seen as expanding the coding capacity of viral genomes, allowing them to access proteomic and functional innovation beyond the confines of their limited genome sizes.

While the effects of NsVGs on viral population fitness remain largely hypothetical or limited to cell culture systems, there is one documented example where the presence of DVGs was associated with enhanced transmission. A 2001 outbreak of dengue virus type 1 (DENV1) in Myanmar revealed the existence of a viral lineage that maintained a nonsense substitution within the E glycoprotein at high frequency (63). This substitution results in the production of a truncated, nonviable E protein and thus should be lethal in the absence of coinfection and complementation with full-length E. Despite this, modeling analysis of transmission dynamics during the outbreak strongly suggested that the presence of the nonsense substitution enhanced DENV1 transmission compared with DENV1 lineages that did not carry it (64). These data provide compelling

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real-world evidence that DVGs can enhance the fitness of viral populations in some circumstances. It remains to be seen whether this is an outlier or whether other examples of NsVGs enhancing the overall fitness of viral populations exist.

Semi-infectious particles, multipartite viruses, and everything in-between. Viruses with segmented genomes inhabit a continuum from monopartite (i.e., all genome segments are copackaged with high efficiency) to multipartite (i.e., each genome segment is separately packaged) (65–67). It has become increasingly clear that even monopartite segmented viruses with selective packaging schemes can produce large numbers of virions that lack functional copies of all genome segments. This phenomenon was first described in depth for influenza viruses (68–71), but it has since been observed during the replication of diverse segmented viruses, including bunyaviruses (72–74) and segmented flaviviruses (75). These virions, termed semi-infectious particles (SIPs), cannot replicate on their own but can still contribute to replication through coinfection, as discussed below.

By varying the actual coding capacity of individual virions, SIP production adds another layer of diversity within many viral populations. For example, variation in the rates at which individual gene segments get packaged into virions by segmented and multipartite viruses can alter the stoichiometry of gene segments within the population, referred to as the genome formula (76, 77). This phenomenon has been best characterized in the context of multipartite plant viruses, where the population frequencies of different gene segments can vary depending on the specific host, resulting in significant changes in overall viral gene expression through gene-dosage effects (77–79). Similar dynamics also have been described for animal viruses, including influenza A virus and bluetongue virus, suggesting that this could be a common feature of segmented viruses (80, 81). Alterations in the genome formula have been hypothesized to facilitate rapid, adaptive changes in gene expression patterns in response to environmental changes (82, 83). This represents a novel form of gene expression regulation that acts at the population level, rather than at the level of individual genomes.

Functional and evolutionary consequences of genomic diversity. Patterns of genomic heterogeneity can significantly affect population-level phenotypes. What are less clear are the evolutionary and ecological forces that produce different forms of nearly ubiquitous genomic heterogeneity across diverse virus families. Several factors may explain the maintenance of genomic heterogeneity, including (*a*) constraints or trade-offs associated with viral replication that make the production of different aberrant genomes and noninfectious particles unavoidable, (*b*) the production of NsVGs, SIPs, and other forms of genomic heterogeneity that may be advantageous and thus maintained by selection (as seems to be the case for the DENV1 example above), and (*c*) the production of NsVGs and SIPs that represents the expected proliferation of so-called cheaters that occurs in biological systems with some degree of cooperation (84). The relative contributions of these (or other) mechanisms likely depend upon viral life history and genome structure.

VIROCELL HETEROGENEITY

Just as no two virions are the same, no two infected cells behave exactly the same. Even when MOI and infection timing are carefully controlled, the patterns and dynamics of viral gene expression vary substantially among infected cells (85–91). Viral burst sizes from individual infected cells span multiple orders of magnitude (85, 86, 92–95). These findings across multiple virus families are consistent with the idea that cell-cell transmission may be dominated by a small proportion of superspreader cells, similar to what is observed at the epidemiological scale (96). Interestingly, at least for influenza viruses, viral burst sizes do not appear to be well-correlated with overall levels

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of viral gene expression, indicating that the drivers of heterogeneity in burst size are likely complex (85). Individual infected cells also exhibit substantial heterogeneity in their transcriptional responses to infection, most notably in the activation of canonical antiviral factors such as type I interferons (87, 89, 97–99). As a result of these patterns, a relatively small fraction of infected cells likely plays a disproportionate role in driving infection dynamics, both in virus reproduction and in the activation of host immune defenses. Thus, characterizing and quantifying these patterns of phenotypic heterogeneity across individual infected cells is crucial for understanding viral infection dynamics and evolution.

The phenotypes of individual infected cells result from the interplay between the viral genetic and genomic variation described above (87, 100) and preinfection variability in host cell states (101–103). Forterre (104) coined the term virocell to refer to infected cells, emphasizing their distinct identity from uninfected cells and virions (which are metabolically inert). The virocell concept provides a useful framework for understanding infection dynamics because it highlights that the behavior of each infected cell reflects the product of unique gene–environment interactions between the viral genome content present in the cell and the state of the cell at the time of infection (**Figure 2**). The phenotypes of individual virocells, in terms of virus production dynamics and cellular infection responses, are where the rubber meets the road for understanding viral genotype-phenotype relationships.

Another important point implicit in the virocell concept is that the combined population of viral genomic elements within a given cell determines the virocell's phenotype. While in some cases this may involve a standard viral genome delivered by a single virion, it also allows diverse genomic content from multiple virions to collectively encode the virocell genome.

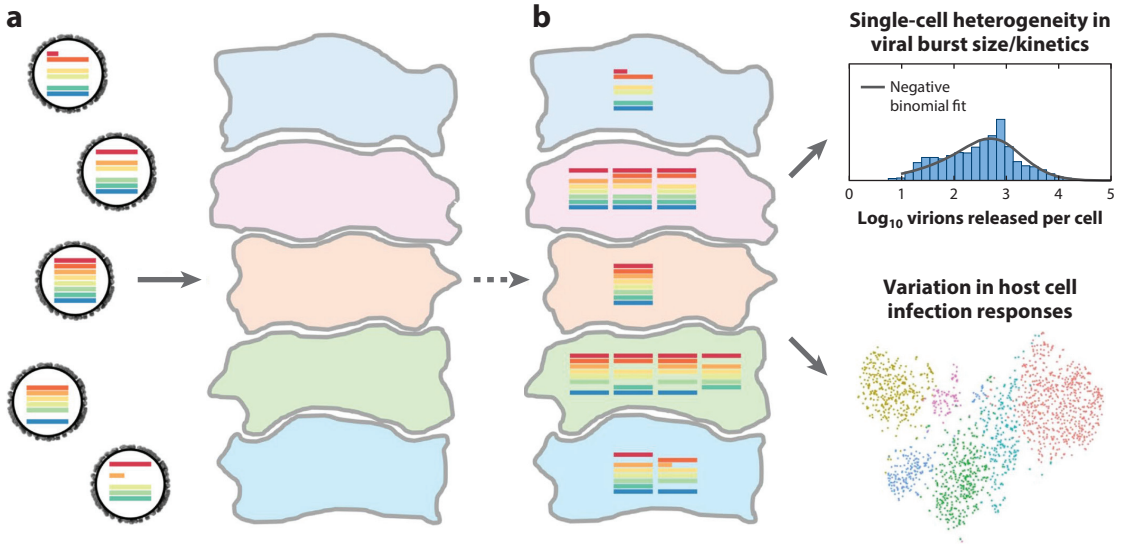


Figure 2

The virocell concept, illustrated with a segmented virus. (*a*) A heterogeneous viral population, harboring substantial genetic and genomic variation, infects a population of cells. These cells exhibit heterogeneity in the preinfection cell state (e.g., gene expression patterns, cell cycle state, signaling state, metabolic state). Per-cell multiplicity of infection varies due to patterns of virion dispersal and coinfection. (*b*) Individual virocells vary in the assemblages of viral gene segments present, as a function of both viral heterogeneity and coinfection dynamics. The interplay between underlying cell state and variation in viral genomic material delivered gives rise to heterogeneity in individual virocell phenotypes such as viral burst size, replication kinetics, and host transcriptional response. Images on the right are adapted from References 87 and 94.

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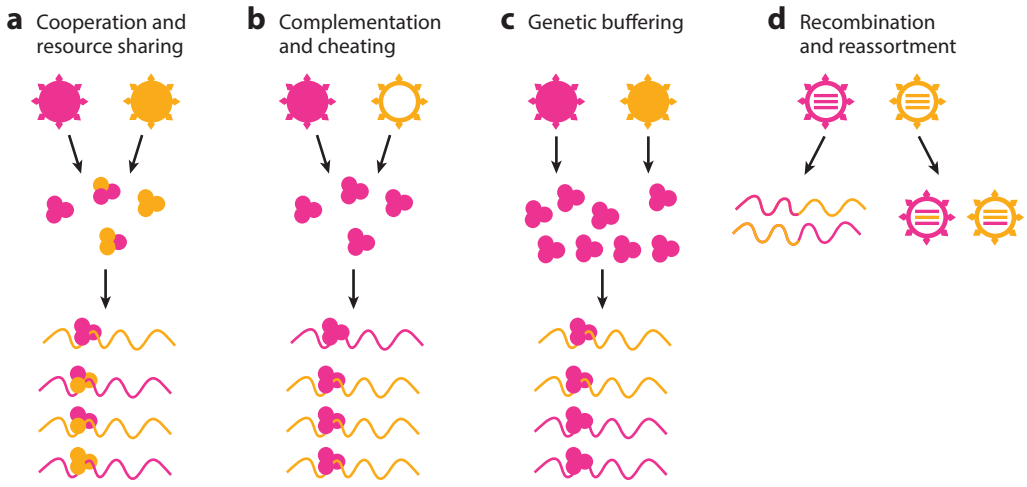


Figure 3

Possible consequences of cellular coinfection. (*a*) Gene products encoded by different parental genomes form mixed, hybrid complexes. In the absence of intracellular compartmentalization, this will result in sharing of phenotypes and fitness between all genomes within the cell. (*b*) Viruses that do not encode essential gene products can have those products provided in *trans* by coinfecting viruses, permitting replication of otherwise nonviable genomes. (*c*) The fitness effects of low-fitness variants (*yellow*) are mitigated by complementation by high-fitness variants (*pink*) and vice versa. (*d*) The production of genetic hybrid progeny can occur through either template-exchange recombination (*left*) or reassortment (*right*).

Under conditions of cellular coinfection, the virocell effectively becomes polyploid (or possibly aneuploid in the case of segmented and multipartite viruses), setting the stage for complex genetic interactions among variable arrangements of viral genomic molecules that govern the phenotypes of individual virocells. The virocell concept argues that it is essential to understand and account for how viral genomic molecules assemble within cells through coinfection and how complex viral genetic interactions within cells influence virocell phenotypes, infection dynamics, and the evolutionary forces acting on viral populations.

Effects of Coinfection on Virocell Behavior and Viral Evolutionary Dynamics

Because the combined viral genomic content delivered to a given cell collectively influences the virocell phenotype, the extent of cellular coinfection can significantly affect infection dynamics. In addition, the complex interactions among individual genomic molecules within a cell (**Figure 3**) can profoundly affect the evolution of viral populations.

Virus-virus interactions, resource sharing, and parasitism. Within individual virocells, viral genomes interact with one another through the sharing of cellular resources, the generation and sharing of viral public goods, and the reshaping of the cellular environment (e.g., stimulation of immune defenses, alterations in cellular metabolism). The interplay between these interactions determines where they fall along a continuum from cooperation and mutualism to parasitism, depending on the net benefit or costs to all involved.

At a basic level, how resources are shared within the cell determines the relationship between cellular MOI and viral productivity (105, 106), which varies among viruses (107–111). For example, VSV replication is more rapid and productive when multiple viral particles are introduced into a single cell (compared to infections in which the same number of particles are spread across multiple cells) (107). Similar outcomes have been observed for polioviruses (112), rotaviruses,

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and noroviruses (113), which induce more efficient infections when multiple virions are delivered together via encapsulation within transmissible vesicles. These instances of positive density dependence likely reflect scenarios in which viruses contribute shared resources [e.g., by enhancing suppression of cellular antiviral defenses through mass action effects (107, 114, 115)] or when cellular resources are not a limiting factor. Under these conditions, viral features that promote cellular coinfection (e.g., virion aggregation or the formation of collective infectious units) are likely to be beneficial.

However, these collective interactions also can support the emergence of parasitism. The large genomic deletions or rearrangements encoded by NsVGs prevent the production of key gene products that could be shared, making NsVGs reliant on public goods produced by coinfecting viruses to sustain their own replication (55). Certain NsVGs also exhibit enhanced immunostimulatory effects relative to WT genomes, which can promote further antagonistic interactions during coinfection (62, 116, 117). Therefore, in some cases, NsVGs can act as cheaters that benefit from coinfection at the expense of their coinfection partners (84).

Beyond NsVGs, genomes that encode gene-lethal or highly deleterious mutations can also function as cheaters, either by acting as a net drain on shared resources or by producing gene products with dominant negative activity that inhibits the replication of other viral genomes within the cell. For viruses with high mutation rates that generate diverse within-host populations, these low-fitness variants are likely to comprise a significant fraction of the population (38, 118). Thus, both NsVGs and low-fitness variants can function as cheaters, raising the overall costs of cellular coinfection for the population.

Genetic complementation and buffering. Coinfection between different viral genetic variants introduces another mechanism by which virus-virus interactions can influence viral evolutionary dynamics. For example, high-fitness variants can mask the deleterious effects of lower-fitness variants during coinfection through genetic buffering, thereby extending the persistence of genetic diversity that might otherwise be lost via purifying selection (119–122). This buffering of fitness effects promotes collective mutational robustness, as individual mutations are expected to have a smaller effect on the virocell's overall viral output when multiple variants are present. Mutational robustness is important for maintaining phenotypic coherence when mutation rates are high (19, 38, 123), but it also can facilitate adaptation by expanding the neutral evolutionary space. This allows populations to sample a larger expanse of sequence space with fewer consequences, increasing the chances of accessing adaptive peaks (19, 124–126) and supporting the retention of variants that are neutral or deleterious in the present context but that may facilitate adaptation in future environments (127–129). Conversely, excessive buffering can be detrimental, as the relaxation of selective pressure on deleterious mutations can lead to population collapse (130).

The strength of genetic buffering effects mediated by coinfection will depend on various factors. First, population density and cellular coinfection levels will determine the opportunity for coinfection-mediated buffering and may change dramatically over the course of infection. Tight within- and between-host transmission bottlenecks will cause cellular MOIs to fall dramatically, eliminating the buffering effect and facilitating the purging of deleterious variants and cheaters as described above. The level of buffering also will depend on the degree of compartmentalization of viral resources within infected cells and the extent to which fitness is determined at the level of the virocell versus the level of individual genome molecules. The advantages of maintaining genetic diversity through buffering also will likely depend on the extent of environmental variability the virus reliably encounters—viruses that regularly encounter environmental variation (e.g., cell types, immune dynamics, host switching) may benefit more from maintaining a larger reserve of variants.

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Collective phenotypes. Interactions between the gene products of variants within coinfected cells can lead to the formation of heteropolymers with new functionalities distinct from those of the parent variants. For measles virus, mixing different F protein alleles in coinfected cells results in the formation of highly fusogenic chimeric F protein trimers that enable direct cell-to-cell spread (131) and expand tissue tropism (132, 133). Similar examples of cooperative interactions also have been observed in clinical cases of measles-associated subacute sclerosing panencephalitis, demonstrating how collective interactions can shape viral pathogenesis (133). In some cases, these interactions can occur between different viral species with compatible tropisms: Experimental coinfections between influenza A virus and respiratory syncytial virus yield chimeric particles with characteristics of both viruses, which demonstrate enhanced antibody escape and expanded receptor usage relative to the parent viruses (134).

Coinfection with multiple variants also permits the accumulation and maintenance of beneficial mutations that are prevented from being combined on the same haplotype by negative epistasis. For instance, experimental infection with populations containing a defined mixture of coxsackieviruses, each encoding different variants, yielded higher viral titers and broader cell tropism than infection with viruses encoding a single genotype in which all variants were encoded together (135). Similarly, populations of influenza A virus harboring a mixture of two alleles at a single site replicated better than clonal populations of either allele alone (136).

Recombination. Recombination also can serve as a mechanism for consolidating beneficial mutations on the same genomes and eliminating lower-fitness hitchhiker mutations. These functions can help reduce clonal interference and promote more efficient selection (137–139). However, recombination also can break up positive genetic linkages (140) or bring together genetic elements that are poorly suited to one another (141). Circumstances that increase the likelihood of genetic mismatch include cases where gene products physically interact with one another (142–144) or where genes perform complementary activities that must be balanced (39, 81, 145, 146). Thus, the scope of biochemical functions encoded by a virus and their distribution across individual genes will play important roles in determining the extent of epistatic interactions within a viral genome and the fitness consequences of recombination.

Both phylogenetic and experimental studies have described the varying fitness effects of viral recombination. Reassortment-mediated recombination is commonly observed among circulating segmented viruses (147–150). Long-term phylogenetic investigations of influenza have shown that lineages that have undergone more reassortment events tend to persist longer (151). Similarly, analysis of recombination rates in seasonal coronaviruses has shown that higher recombination levels are associated with increased adaptability (152). Within hosts, recombination between polioviruses is associated with expansion into new tissue sites (153) and also may be necessary for the efficient removal of deleterious mutations arising from rapid replication (13). Recombination also can rescue nonfunctional viruses by swapping genetic material with cocirculating functional viruses. Of particular interest is the restoration of virulence to live attenuated vaccine strains following recombination with circulating viruses, as has been periodically observed with the Sabin live attenuated polio vaccine (154, 155), although more recent updates to the vaccine have helped curb the capacity for recombination (156).

In other cases, recombination is associated with negative fitness outcomes. Experimental rescue attempts of various reassortant H1N1-H3N2 combinations revealed that many of the resulting viruses were either attenuated or nonviable (157). Consistent with these observations, the rate of detectable reassortment events between seasonal influenza viruses is lower than would be expected if reassortment yielded only neutral fitness effects, and reassortment rates decrease with increasing genetic distance between strains (158, 159). These findings suggest that the consequences of

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negative epistatic interactions can often outweigh the benefits of antigenic diversification. Additionally, the template-switching mechanism of recombination can generate NsVGs with deletions if the polymerase skips a stretch of the genome before resuming replication (51, 160, 161). As detailed above, these NsVGs can suppress WT replication and induce an immune response that curtails infection (161).

Factors Influencing the Prevalence of Virus-Virus Interactions

While collective interactions between coinfecting viruses are an important force shaping viral dynamics, their relevance in any given infection will depend on the likelihood of viruses encountering one another in the first place. Barriers at the population, host, and cell levels can prevent viruses from interacting during infection. Population-level barriers can arise from geographic differences in strain circulation, restricted interhost interactions, or incompatibilities in host tropism (162–164). Transmission bottlenecks are another mechanism that can lead to interaction barriers, as the diversity of viruses present in each individual is initially limited by the number of genomes that successfully establish infection. For human respiratory viruses (such as influenza and SARS-CoV-2), bottlenecks are generally very narrow, and new infections are founded on only a few genomes (165, 166). However, restrictive transmission is not a general rule—some viruses can be transmitted collectively in clusters or vesicles that contain tens to thousands of viral particles, resulting in broader bottlenecks and greater population diversity at the initiation of infection (111–113, 167).

Bottlenecks also can occur at the within-host level. Tissue-level bottlenecks can arise from physical barriers that funnel virions into separate pockets of replication, a lack of susceptible/permissive cells at a tissue site, or immune restriction. For example, mosquitoes infected by arboviruses often experience a loss of viral diversity in the midgut, a phenomenon that can constrain the evolution of the population at large (168). For mammalian influenza infections, viruses tend to separate into discrete subpopulations in the lower airways, despite being well-mixed in the trachea; this trend is partially attributable to the anatomical differences between these sites (34, 169). The organization of receptor types in the airways also can limit influenza interactions, with human-origin and avian-origin strains preferring separate receptors with distinct spatial distributions (170). Immune barriers at certain tissue sites also can constrain viral distribution. For example, expansion of polio infections to distal tissue sites is checked by the type I interferon response (171), and only certain variants evolve the ability to circumvent this restriction (153).

Even when viruses infect the same tissue site or cell, smaller-scale constraints can limit their interactions. One mechanism of constraint is superinfection exclusion, a virus-mediated process that restricts secondary infections from establishing in preinfected cells. Exclusion can arise from diverse mechanisms across a wide range of viral families (172–176), but it generally blocks infection before genome replication occurs, thereby obstructing interactions such as cooperation, cheating, complementation, and recombination (172). Superinfection exclusion can thereby result in the establishment of a patchwork infection pattern, with genetically distinct clusters of viruses replicating independently of one another (35, 177). Viruses that do manage to coinfect the same cell also may be subjected to intracellular bottlenecks arising from genome degradation or stochastic failure at different stages of the life cycle, resulting in further losses of diversity (95, 178). Certain viruses also undergo within-cell compartmentalization, sequestering resources in replication factories or inclusion bodies and potentially circumventing interactions with coinfection partners (179, 180). Therefore, even high viral density within a host may not guarantee significant virus-virus interactions.

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CONCLUDING REMARKS AND FUTURE DIRECTIONS

An appreciation of viral population diversity is necessary for a complete understanding of infection biology. Across all scales of infection, viral populations are dynamic collections of genetic and genomic variants, and interactions among these variants influence infection outcomes and drive viral evolution. Common features of viral life cycles, such as high mutation rates and recombination, generate heterogeneity and yield populations in which many or most individuals are in some way distinct from the rest of the group. Members of these viral ensembles operate in proximity to one another, infecting the same hosts, tissues, or cells and engaging in a spectrum of cooperative and competitive interactions. The extreme diversity of these populations, coupled with the ubiquity of interactions such as genetic complementation, buffering, and exchange, makes it impossible to describe infections in terms of a single dominant genotype or even multiple genotypes acting independently. Instead, we must understand viral populations as dynamic networks of diverse genomic subunits that are constantly being rearranged into new configurations within individual virocells. The structure of these dynamic networks is determined by patterns of cellular coinfection and mixing. This conceptualization enables a deeper understanding of infection biology and points toward further research directions into the ecology and evolution of viruses.

New research approaches and conceptual frameworks are needed to better understand the implications of population diversity and collective interactions for viral populations. One major unresolved question concerns the role of NsVGs and incomplete particles in infection. The mechanisms governing their generation are undefined for most viral species, and their evolutionary situation is even less clear. Do NsVGs exist as purely selfish elements that inevitably emerge due to the inherent advantages of a cheater lifestyle? Or does their presence confer some population-level benefit that results in their maintenance across a multitude of viral systems? Does the prevalence of incomplete particles yield a reproducible genome formula, and if so, is this formula able to reliably shift to adapt to selective pressures? Why do certain segmented viruses produce incomplete particles while others do not?

The level of organization at which selection acts during infection also requires further consideration and investigation. Although fitness is often discussed in terms of individual genotypes, the widespread sharing of viral gene products means that many fitness effects are experienced collectively by the entire population within a cell, suggesting that selection generally acts on the virocell. Additionally, when population structuring is strong within a tissue, altruistic interactions can arise between infected cells in the same cluster (181). Group selection acting on multiple virocells within a focus of infection must be considered. In this case, viral and host factors that influence coinfection dynamics and the distribution of genomic molecules across virocells would be expected to be important determinants of population-level phenotypes and fitness. Further research will be necessary to determine the relevance of these levels of organization for viral evolution and the extent to which fitness interests are aligned or divergent across scales.

Finally, future research may focus on developing a deeper understanding of how viral interactions and cellular-level population dynamics can shape long-term evolution. For example, the diversity generated by mutation and recombination may often prove beneficial in the long run (due to changing environments and novel pressures) but can be deleterious to maintain in the short term. How do viruses balance these opposing interests? To what extent does transmission between individuals (or even between different tissue sites within an individual) represent an environmental shift that requires a response at the genotypic level? Is viral plasticity during infection mediated more by mutation or by nonmutational changes in gene expression? It also is unclear how long diversity can be preserved at low levels before expanding in response to a changing fitness landscape. Observation of global-scale evolutionary dynamics suggests that variants can persist at

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low frequencies for extended periods of time before re-establishing. For example, the SARS-CoV-2 Omicron lineage emerged from the Alpha lineage, which at the time had been almost completely supplanted by Delta (182). Diversity-preserving viral interactions (such as complementation and buffering) are therefore likely to be important forces governing strain dynamics at higher levels.

Addressing these questions will deepen our mechanistic understanding of evolutionary dynamics across diverse viral taxa and host systems and will inform the development of more effective control strategies for rapidly evolving pathogens.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

We gratefully acknowledge the support of National Institutes of Health grants R01AI139246, R01AI179910, U01AI186993, and U19AI181767.

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